

CBCS SCHEME

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Third Semester B.E. Degree Examination, June/July 2024 Digital System Design using Verilog

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. The input to a combinational logic circuit is a valid single digit BCD data. Design the logic circuit using minimum hardware to detect whenever a number greater than 5 appears at the input. (08 Marks)
- b. Expand $f_1 = a + ab + a\bar{c}d$ into minterm and $f_2 = a.(b + c).(a + c + \bar{d})$ into maxterm. (06 Marks)
- c. Reduce the following function using K-map technique and implement using gates.
 $f(P, Q, R, S) = \sum m(0, 1, 4, 8, 9, 10) + \sum d(2, 11)$ (06 Marks)

OR

- 2 a. Identify all prime implicants and essential prime implicants of the following function using K-map.
 $f(a, b, c, d) = \sum m(0, 1, 2, 5, 6, 7, 8, 9, 10, 13, 14, 15)$ (06 Marks)
- b. Simplify the following Boolean function using Quine McCluskey method.
 $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$ (10 Marks)
- c. Minimize the expression using K-map
 $Y = (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + B + C) \cdot (A + B + C)$ (04 Marks)

Module-2

- 3 a. Design 2-bit magnitude comparator. (10 Marks)
- b. Implement $f(w, x, y, z) = \sum m(0, 4, 8, 10, 14, 15)$ using
 - i) 8×1 MUX with w, x, y as select lines.
 - ii) 4×1 MUX with w, x as select lines.(06 Marks)
- c. Implement full adder using 74138 decoder. (04 Marks)

OR

- 4 a. Explain the general structure of PLDs. (06 Marks)
- b. Construct 4 to 16 line decoder from 2 to 4 line decoder and implement the Boolean function.
 $f(x_3, x_2, x_1, x_0) = \sum m(0, 6, 9, 11, 15)$ (08 Marks)
- c. Design 3 bit binary full subtractor using logic gates. (06 Marks)

Module-3

- 5 a. With neat diagram explain Master Slave JK Flip Flop. (08 Marks)
- b. Explain 4 bit universal shift register. (08 Marks)
- c. Obtain the characteristics equation for SR and T Flip Flop. (04 Marks)

OR

- 6 a. Explain the working of Mod-4 twisted ring counter. (07 Marks)
- b. Design 4 bit binary ripple counter using T Flip Flop. (06 Marks)
- c. Design Mod-6 synchronous counter using clocked JK Flip Flop. (07 Marks)

Module-4

- 7 a. With general syntax and suitable examples, explain the shift operators available in verilog. (06 Marks)
 b. List and explain the verilog data types. (08 Marks)
 c. Realize full adder circuit using verilog data flow description. (06 Marks)

OR

- 8 a. Write a verilog code for a 2×1 multiplexer in dataflow description using signal assignment. (06 Marks)
 b. List all the different styles of descriptions, explain the structure of dataflow description. (06 Marks)
 c. Explain the following in data flow description:
 i) Signal declaration and assignment statement
 ii) Constant declaration and assignment statement
 iii) Concurrent signal assignment statement
 iv) Assigning a delay time to the signal assignment statement. (08 Marks)

Module-5

- 9 a. Explain CASE statement with syntax. Write a behavioral description of a positive edge triggered JK Flip Flop using CASE statement in verilog. (10 Marks)
 b. Write verilog behavioral description of 8×1 MUX. (06 Marks)
 c. Write a verilog program for half adder using structural description. (04 Marks)

OR

- 10 a. With example, explain the syntax of following sequential statements:
 i) If ii) Else if (06 Marks)
 b. List and explain all the loop statements in verilog. (08 Marks)
 c. Write a verilog program for 3 bit binary counter using case statement. (06 Marks)

Model Question Paper-I with effect from 2023-24 (CBCS Scheme)

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Third Semester B.E. Degree Examination Digital System Design using Verilog

TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	Design a logic circuit that has 4 inputs, the output will only be high, when the majority of the inputs are high. Use k-map to simplify	L3	10
	b	Simplify the given Boolean function using Quine MC Cluskey find the prime and essential prime implicants and also verify with K-map $f(A,B,C,D)=\sum m(0,1,4,5,7,8,13,15)+d(2)$	L3	10
OR				
Q.02	a	Simplify the following expression using K-map. Implement the simplified expression using basic gates only $f(a,b,c,d)=\prod m(0,2,3,4,5,12,13)+dc(8,10)$	L3	10
	b	Define the following terms with example. Minterm, Maxterm	L1	04
	c	Place the following equation into proper canonical form $P = f(a,b,c) = ab'+bc$ $T = f(a,b,c) = (a+b)(b'+c)$	L3	06
Module-2				
Q. 03	a	Design two bit magnitude comparator and write truth table, relevant expression and logic diagram.	L3	08
	b	Implement the following functions using 3:8 decoder $f_1(a,b,c) = \sum m(1,3,5)$ $f_2(a,b,c) = \sum m(0,1,6)$	L3	6
	c	Implement $Y=ad+bc'+bd$ using 4:1 mux considering A and B as a select line.	L3	6
OR				
Q.04	a	Explain 4-bit carry look ahead adder with neat diagram and relevant expressions.	L2	10
	b	Implement the following Boolean function using 8:1 multiplexer and 4:1 multiplexer $f(a,b,c,d) = \sum m(0,1,5,6,10,12,14,15)$.	L3	10
Module-3				
Q. 05	a	Explain the working of Master-Slave JK flip-flop with functional table and timing diagram.	L2	10
	b	Explain Universal Shift Register with the help of logic diagram, mode control table.	L2	10
OR				
Q. 06	a	Derive Characteristic equation for SR,T,D and JK flip-flop with the help of function table	L2	10
	b	Design a Synchronous Mod-6 counter using SR flip-flop for the sequence 0-2-3-6-5-1.	L3	10

Module-4				
Q. 07	a	Realize the 2x1 Multiplexer with active low enable and also write the Verilog program by considering delay time to the signal assignment statements with simulation waveform.	L3	10
	b	Realize the Full Subtractor circuit using Verilog data flow description.	L3	10
OR				
Q. 08	a	List all the data types available in Verilog HDL and explain any four data types with examples.	L2	10
	b	Explain arithmetic and logical operator with example.	L2	10
Module-5				
Q. 09	a	Realize the D-Latch Behavioral description code with circuit diagram, waveform.	L3	10
	b	Write a Verilog structural code for four bit ripple carry adder.	L3	10
OR				
Q. 10	a	Explain with syntax of the following sequential statements in Verilog. i) For-loop ii) While-loop iii) Repeat iv) Forever	L2	8
	b	Write a Verilog program for 8:1 MUX using case statement.	L3	6
	c	Realize the JK flip-flop using Verilog Behavioral description.	L3	6

Model Question Paper-II with effect from 2023-24 (CBCS Scheme)

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Third Semester B.E. Degree Examination Digital System Design using Verilog

TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	What is combinational circuit? Design a combinational logic circuit with three input variables that will produce logic 1 output when more than one input variables are logic 1.	L2	06
	b	Convert the following Boolean function into canonical minterm and maxterm form in decimal format $R = f(a, b, c) = a + b(a + c) + bc$	L2	06
	c	Find all the prime implicants and essential prime implicants for the following function using k-map method. (i) $M = f(a, b, c, d) = \Sigma(1,5,7,8,9,10,11,13,15)$ (ii) $Y = f(a, b, c, d) = \pi(0,2,3,8,9,10,12,14)$	L3	08
OR				
Q.02	a	. Simplify the following Boolean functions using K-map (i) $Y = f(a, b, c, d) = \pi(0,1,4,5,8,9,11) + d(2,10)$ (ii) $M = f(w, x, y, z) = \Sigma(0,1,2,4,5,6,8,9,12,13,14)$	L3	10
	b	Solve the following Boolean function by using QM minimization technique $P = f(w, x, y, z) = \Sigma(2,3,4,5,13,15) + d(8,9,10,11)$ Verify using K-map method	L3	10
Module-2				
Q.03	a	Implement the following functions using 3:8 decoder along with OR and/or NOR gates. In each case the gates should be selected so as to minimize their total number of inputs. (a) $f_1(X_2, X_1, X_0) = \Sigma m(0,2,4,6,7)$ and $f_2(X_2, X_1, X_0) = \Sigma m(1,3,5,6,7)$ (b) $f_1(X_2, X_1, X_0) = \Pi M(0,2,4,6,7)$ and $f_1(X_2, X_1, X_0) = \Pi M(1,3,7)$	L3	07
	b	Construct single decade decimal adder with necessary correction circuit design.	L2	08
	c	Design a one-bit comparator circuit.	L2	05
OR				
Q.04	a	Construct the following function $S = f(a,b,c,d) = \Sigma(1,4,5,7,8,9,14,15)$ using (a)8:1 Mux and (b)16:1 Mux	L2	07
	b	Construct the functional table for 4 to 2 line priority encoder with a valid output, assigning highest priority to highest bit position or input with highest index and obtain the minimal sum expressions for the outputs.	L2	07
	c	Implement a Full adder using PAL	L2	06
Module-3				
Q.05	a	Explain the working of Master-Slave JK flip-flop with functional table and timing diagram	L2	08
	b	Derive the characteristic equations of SR and D Flipflops	L2	04
	c	Make use of negative edge triggered T Flip Flops to describe the working	L2	08

		of 4-bit binary ripple counter. Also draw the timing diagram.		
		OR		
Q. 06	a	Make use of 4-bit shift register circuit to explain the following modes of operations: SISO, SIPO, PISO and PIPO.	L2	08
	b	Develop a mod-5 synchronous counter with the sequence 0,2,6,3,1 using T-Flip Flops.	L3	07
	c	Explain the working of Ring counter with necessary diagram and equations.	L2	05
		Module-4		
Q. 07	a	List all the data types available in Verilog HDL. Explain any three data types with examples.	L2	08
	b	Explain three modeling styles available in Verilog with half adder example	L2	07
	c	(ii) Evaluate the following: i) $A * B$ ii) $A + B$ iii) $A << 2$ iv) $\{A[3], B\}$ Given: $A=0011$ $B=0100$	L2	05
		OR		
Q. 08	a	Explain the Signal declaration and Assignment Statements in Verilog data-flow description.	L1	06
	b	Realize D latch with active high enable and also write the Verilog program by considering delay time to the signal assignment statements with simulation waveforms.	L2	08
	c	Write a verilog dataflow model for full adder	L2	06
		Module-5		
Q. 09	a	Write a verilog behavioural description for 8:1 mux along with the design and timing diagrams.	L2	08
	b	Explain different case statements available in verilog with syntax and necessary examples	L2	06
	c	Realize the Binary up-down counter using verilog behavioral description.	L2	06
		OR		
Q. 10	a	Explain with syntax of the following sequential statements in Verilog. i) For-loop ii) While-loop iii) Repeat	L2	06
	b	Write a structural description of 3-bit ripple carry adder	L2	08
	c	Write a verilog code for 2x1 mux using If Else statement	L2	06